From a grasp pose to a planned motion

MoveIt 2 combines inverse kinematics, joint-space search and Cartesian path computation.

Inverse kinematics

POSE → JOINT CONFIGURATIONS

Target: position + orientation

$$FK(q) = T_target$$

$$q = [q_1, q_2, \ldots, q_6]$$

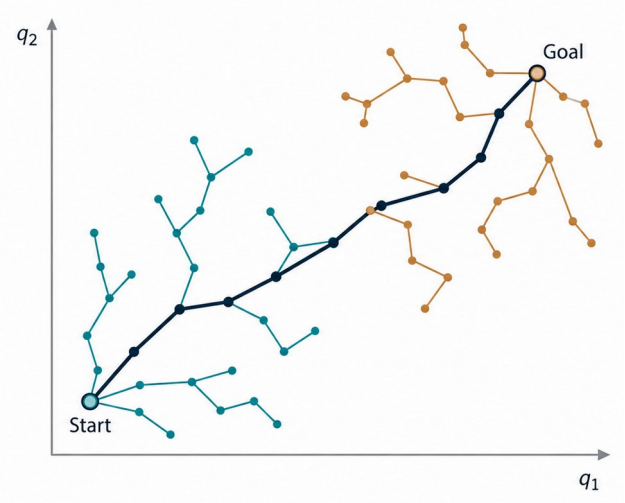
One tool pose, multiple IK solutions.

Rank by $J = \Sigma_i(q_i - q_{i,current})^2$

8 IK attempts; plan up to 3 solutions.

RRTConnect / OMPL

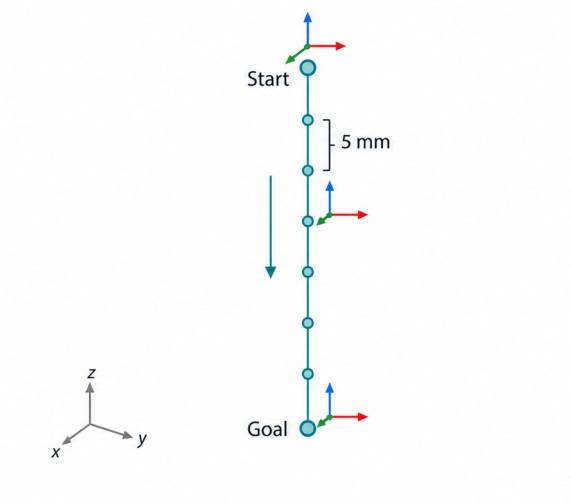
BIDIRECTIONAL SAMPLING



Grow trees from start and goal.
Connect valid joint states.

Cartesian interpolation

TOOL-SPACE WAYPOINTS



Line waypoints → successive IK.
5 mm step; full path required.

Method illustrations: joint-space projection (centre), fixed-orientation tool path (right). Sources: M1, M3.

Plan the complete task, then validate and execute

Two joint-space transfers and four Cartesian segments are planned before any physical motion.

01 Approach	02 Descend	03 Lift	04 Transfer	05 Lower	06 Return
10 cm above rim	To grasp pose	+10 cm	Fixed ready pose	−20 cm	+20 cm
RRTConnect	**Cartesian**	**Cartesian**	**RRTConnect**	**Cartesian**	**Cartesian**

Gripper sequence: open before descent → close before lift → hold still for 1 s at drop → release → return.

01 STATE & GEOMETRY

Valid IK and workspace
Joint bounds along the path
Sampled scene collision checks

02 CARTESIAN FIDELITY

Complete path; no large joint jumps
Bounded lateral / angular deviation
Endpoint: 2 mm and 0.02 rad

03 TIMING & EXECUTION

Scale time to speed / acceleration limits
Reject excessive motion duration
Recheck start state and controller

PRE-MOTION SELECTION Try another IK solution; if infeasible, try the next grasp candidate.

Timed joint trajectory → MoveIt ExecuteTrajectory → Robot trajectory controller